



LAND OF THE CURIOUS

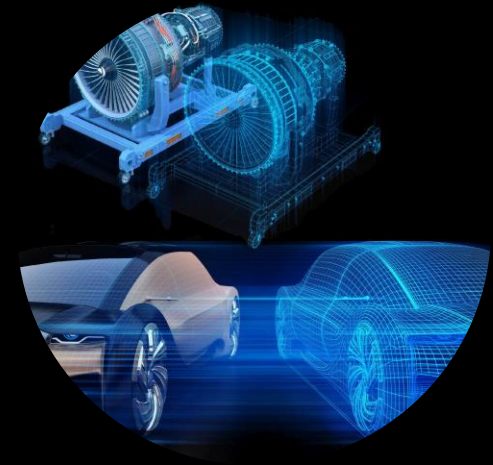


EVENT AND DATE

LECTURE 3 – DECISION MAKING PROBLEM FOR MANAGEMENT

CS39A0040 Product & Service Development

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AGENDA FOR LECTURE

- »» Product Development - The Big Picture & The Management View
 - »» Strategic Alignment
 - »» Portfolio Decisions
 - »» The Reality
- »» What is our Product?
 - »» Product vs. Business model
 - »» Productization level
 - »» Market Segment Selection – The Tough Selection

”

” Defining and understanding your data model is the foundation for AI and Digital Twin implementation ”

Quote from a meeting with Aalto



What is a Technology Strategy?



PRODUCT DEVELOPMENT - THE BIG PICTURE

The management view



IMPORTANCE OF KEY CONCEPTS AND DEFINITIONS

- » Business management is based on concepts and shared understanding of definition between human beings
- » From concrete to abstract, different abstraction levels: the higher in the hierarchy more abstract are concepts-> difficulty to have shared understanding.
- » Definitions are not typically universal but rather context specific (industry, discipline or function, company, country, etc.)
- » *Summary: Words matter but you had to know the context*

PRODUCT ROLE AS AN ENTITY IN BUSINESS

Strategy process

Product Portfolio management

Product
development and
lifecycle mgmt.

Sales &
marketing

Manufacturing
&
Procurement

Delivery &
commissioning

Support

Customer

Installed base (Asset)/Agreements

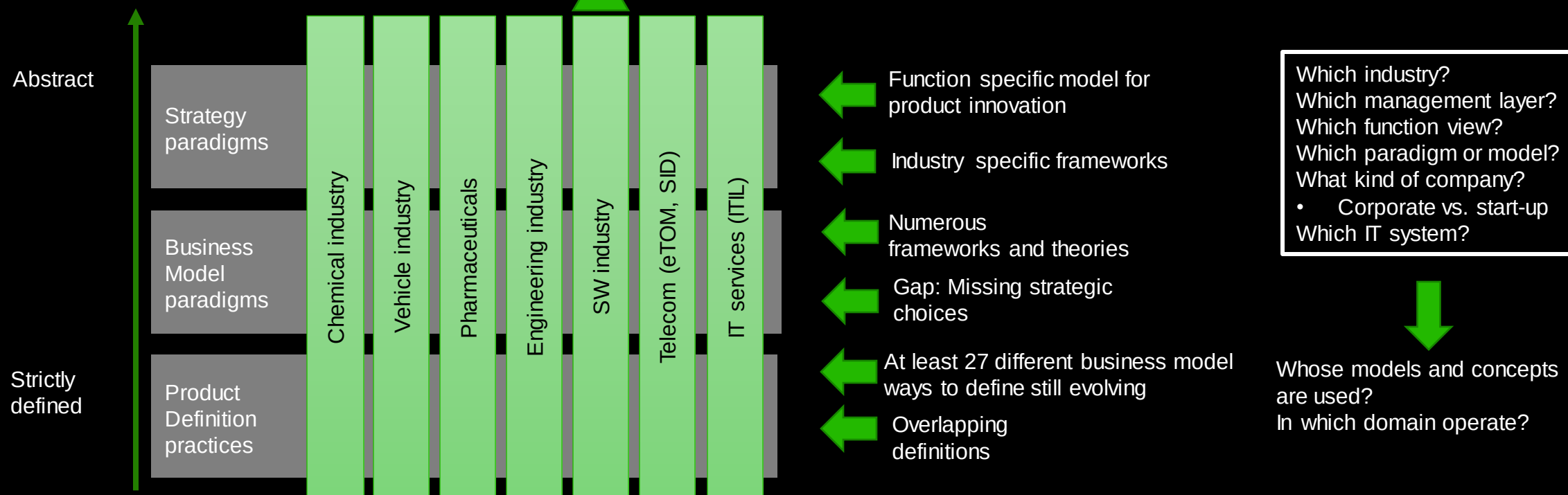
Product

Vendor

Finance

Human Relationship (Employee, customer, supplier)

THE PROBLEM OF CONTEXT SPECIFICITY



- Closing the gaps and aligning & mapping definitions between layers and models is challenging
- However, if you can't model it you can't manage it



SUMMARY

- » Words matter
- » Product development and product definitions are challenging due cross domain nature: both vertically and horizontally
- » There are management paradigms are challenging due to context specificity. The big picture is not always that clear and there are holes.

CHOOSE THE VIEW

Management view

Strategic alignment

Portfolio decisions

Revenue/Profit allocation

Increasing Complexity

Reality: Examples

Product management view

What is our product?

What kind of product?

For what purpose?

How to monetize?

“Product”

- » Target: how product development linked with rest of the business
- » Focus on what rather than how: more generic than specific to certain product type, pain point



I STRATEGIC ALIGNMENT

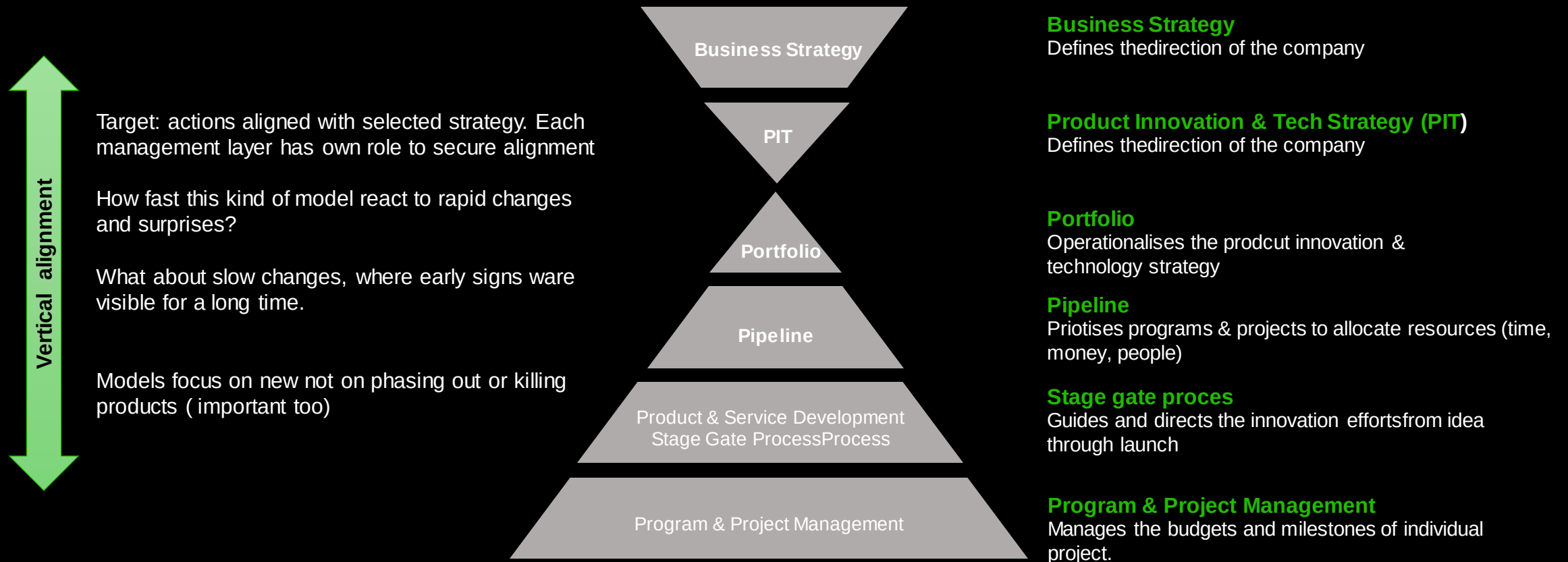
The management view

FRAMEWORK: AN EXAMPLE BY BAIN & COMPANY

Management capabilities	Portfolio management and finance	M&A, JVs and partnering	Regulatory management	BU strategy and driving priorities	HR management and culture
Operating capabilities	Supply chain and logistics	Production and operations	Development and innovation	Go-to-market	Customer relationships
Proprietary asset	Tangible asset	Scale	Technology and IP	Brand	Tied customer network

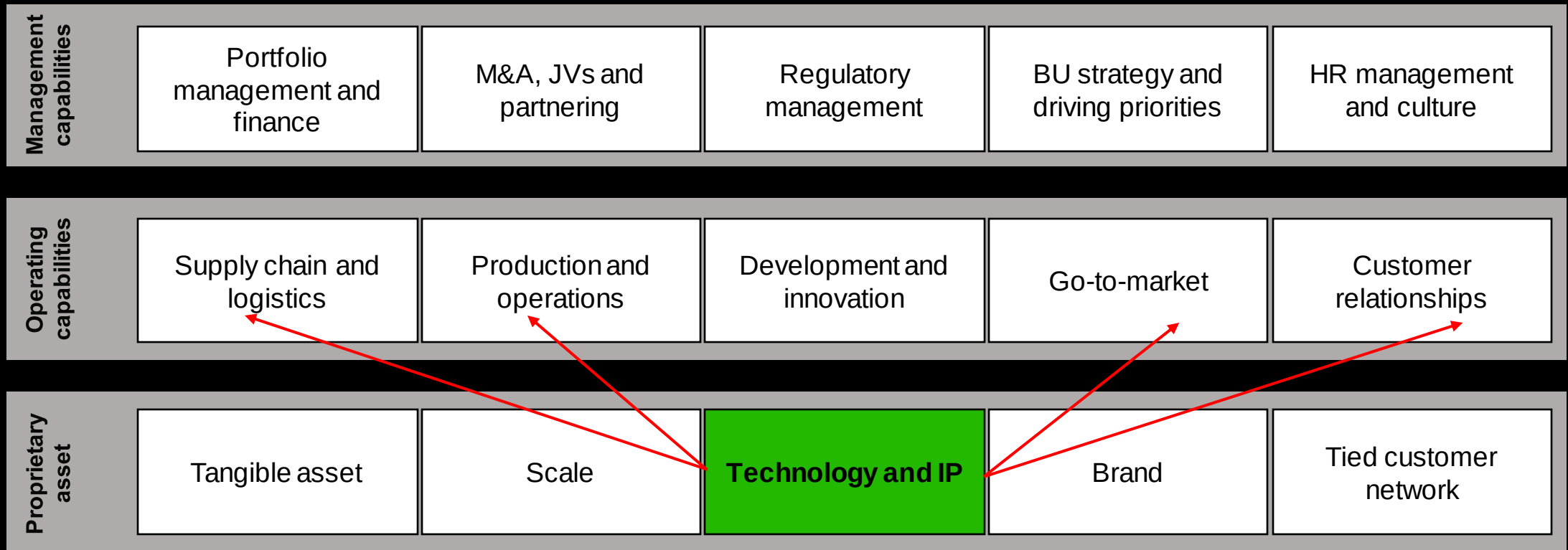
» Where is product and product development? Are they source of differentiation?

PRODUCT INNOVATION: FUNCTION VIEW FOR ALIGNMENT



HORIZONTAL ALIGNMENT: WILL NEW PRODUCTS REQUIRE NEW CAPABILITIES ?

- Misalignment with operational capabilities will lead to poorer performance no matter how good your product is
- Management view: How to differentiate?



DIFFERENT APPROACHES TO PRODUCT INVESTMENTS

Internal development

Own development

Strategic decision to keep inhouse

External partner

We don't have resources or competence

Development with
customer

We need access to real requirements and test environment
Exclusivity to technology given to customer (years)

**Business model variation
in development per product**

Risk & cost sharing with
competitor

Huge cost of platform development thus sharing
Typical in car industry: VW/Ford, Nissan/Renault

License a technology

ARM technology and Apple M1, M2 chips

Buy a technology

Speed is essential, research and new development is not our core
Making acquisitions is our competence



SUMMARY: STRATEGIC ALIGNMENT

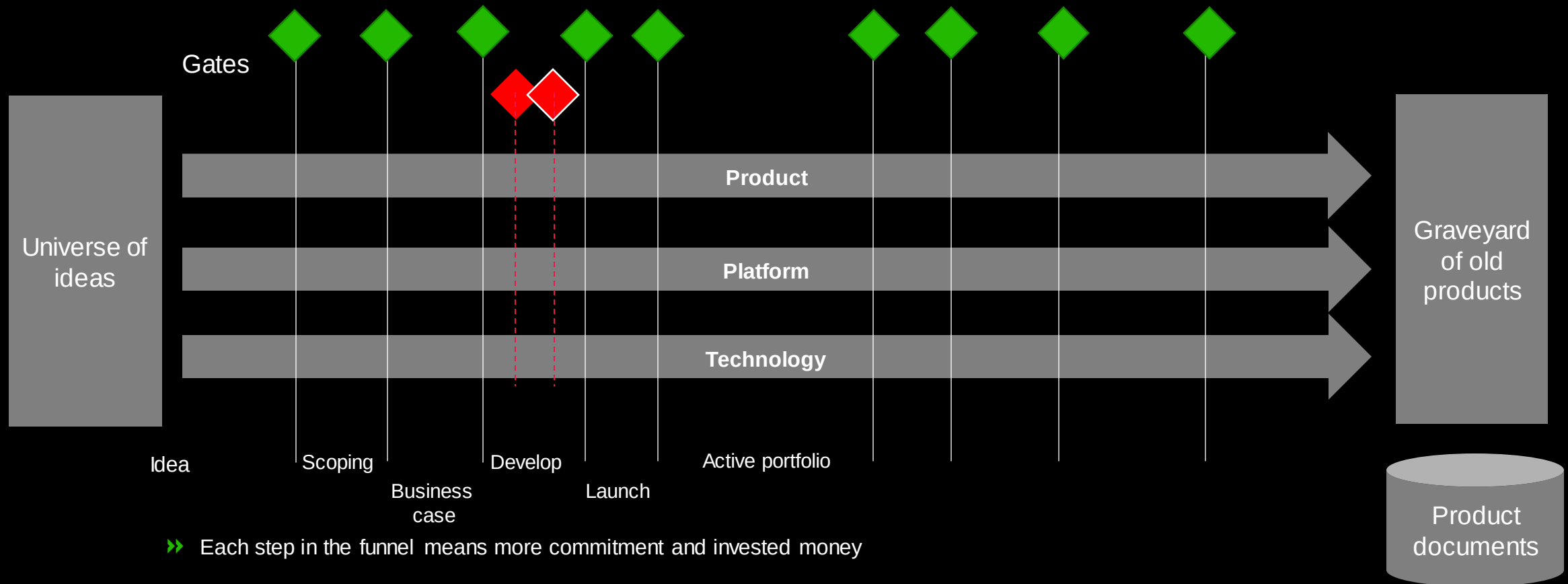
1. What is the role of products and product development in our strategy: is it something that differentiate us?
2. Management sees a product an asset
3. It is important to take care both vertical and horizontal alignment
4. There are multiple ways to acquire the asset each presenting own business model



PORTFOLIO DECISIONS

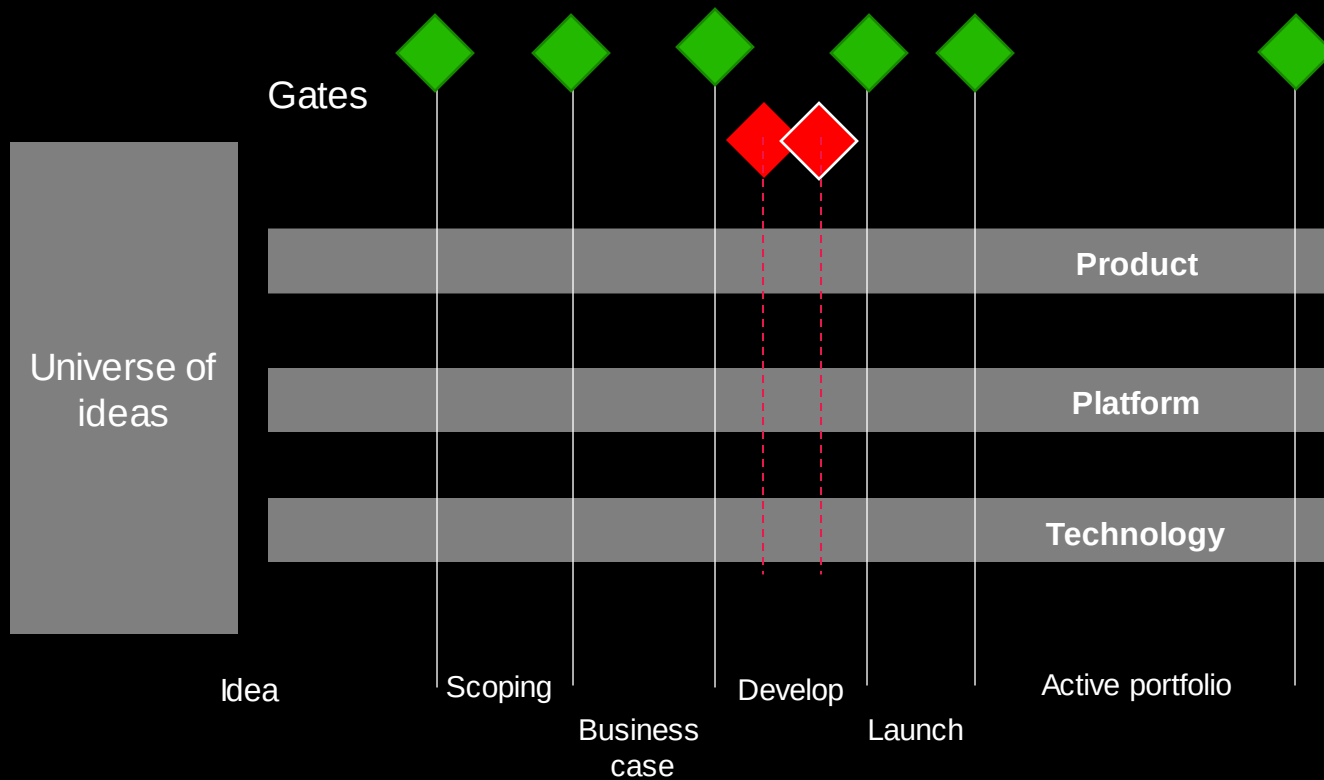
Product is an asset in management frameworks

PORTFOLIO FUNNEL AND ALLOCATION DECISIONS



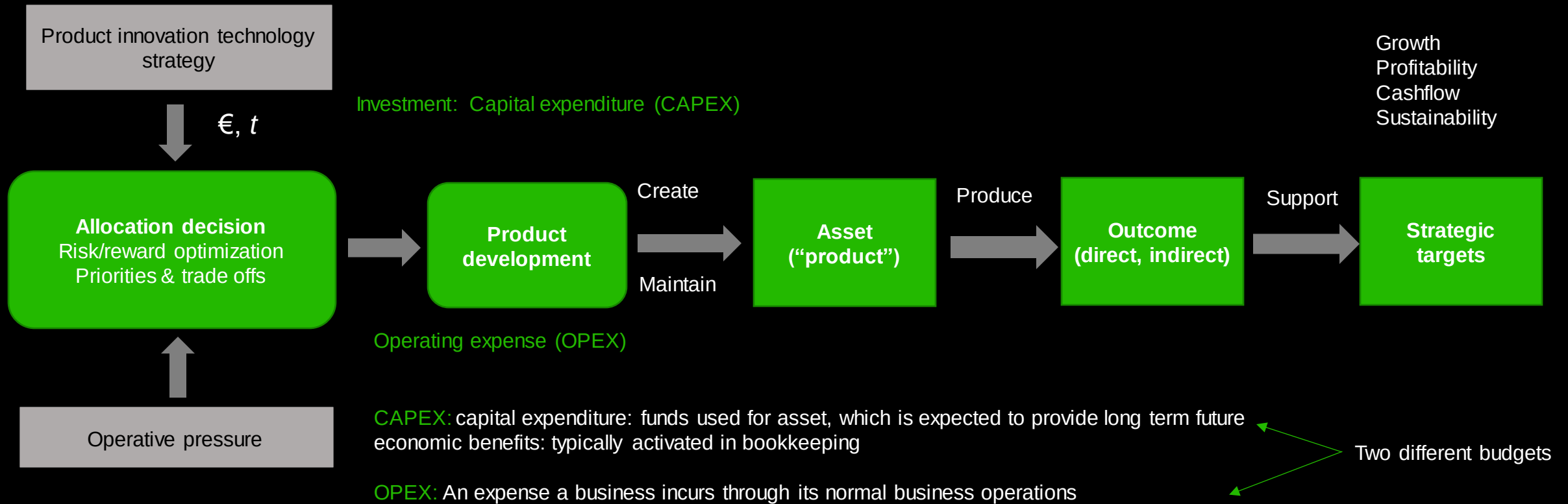
- » Each step in the funnel means more commitment and invested money
- » Gate: **Go/Kill** decisions based on evaluation -> stage, i.e. stage-gate model

ROLE OF GATES



- The role of gate decisions is to screen proposals/ideas, this is like the Idols competition
- Each following step means, more work -> more expensive step and more money invested -> resource allocation decision.
- After each step decision makers know more and ready to allocate or deny more resources (+ hold, re-route)
- Stepwise approach is needed to evaluate the risk and reward equation.
- In general, all industrial investments follow this stepwise approach.
- The name of the phases are different in each investment type
- Documentation and level of detail of information required for decision making is increasing step by step.

ALLOCATION AND ASSET: THE LOGIC

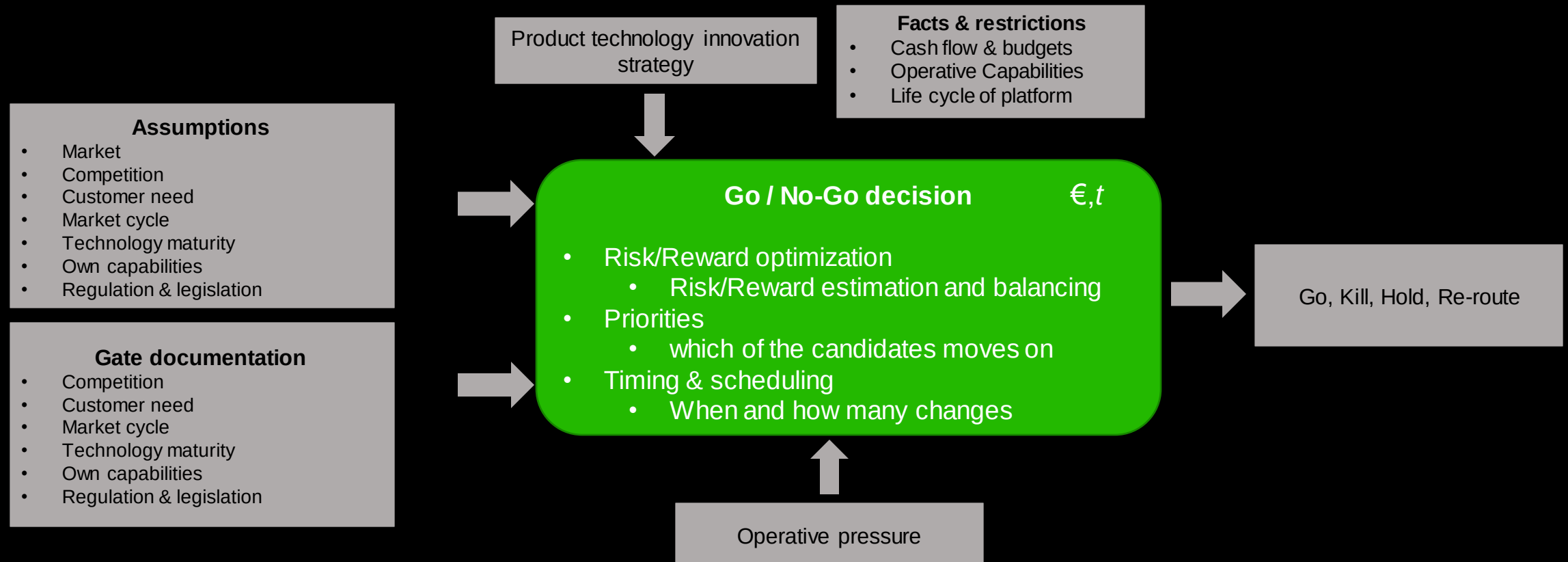




OUTCOME

- » Expected outcome depends on the intended role of the product in the portfolio
- » In case of a new product:
 - » Revenue and profit
 - » Access to a customer segment
 - » Make a foothold where to expand
 - » Create installed base -> life cycle services
 - » Support existing products
 - » Keep competitor away
 - » Etc.
- » In case of improvements:
 - » Cost competitiveness
 - » Sustainability
 - » Quality improvement
 - » Functionality enhancements
 - » Comply legislation or regulation

GATE: RESOURCE ALLOCATION DECISION



» Resource and time allocation are the key methods to secure strategic alignment

THE TRUE NATURE OF ALLOCATION DECISION

Allocation decision $\text{€}, t$

- Risk/Reward optimization
 - Risk/Reward estimation and balancing
- Priorities
 - which of the candidates moves on
- Timing & scheduling
 - When and how many changes

Conflicts due to:

- Different situational awareness
- Strategy interpretations
- Strategic vs. operative pressures
- Beliefs, assumptions
- Personal, function priorities etc.

Conflicts are not necessary a bad thing, but they require **conflict management** process to avoid **confrontation**

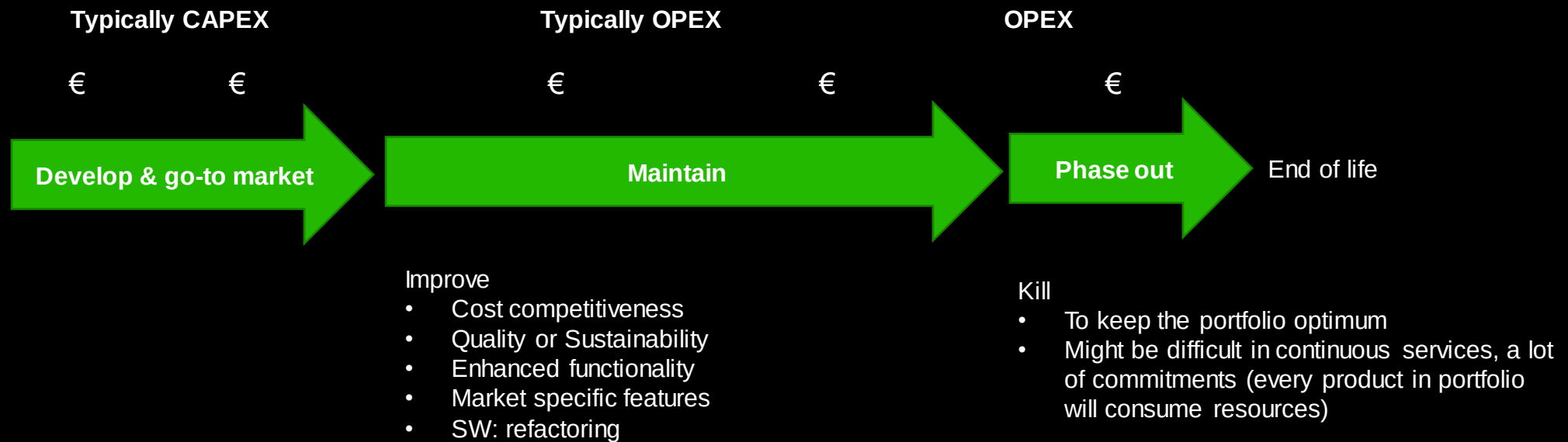
Trade-off decision:

- to gain something you give up something
- Should not be internal politics but sometimes hard to avoid

» *Resource and time allocation are the key methods to secure strategic alignment*

INVESTMENT AND PRODUCT LIFECYCLE

» *Multiple allocation decisions over life cycle*





PHASE OUT : RETIRE PRODUCT

- » Not always so simple due to:
 - » Commitments to customer
 - » Regulation/legislation
 - » High migration cost (old products-> new products)-> bloated portfolio -> hidden cost
 - » KPI and incentives: you get what you measure, especially in sales organization
- » Risk management important
 - » In case of accidents where are you proofs?
 - Design and decision documents archived-> personal risks
 - For long life cycle product this is challenging



SUMMARY: PORTFOLIO DECISIONS

- » Portfolio of product is managed with funneling development ideas through gates and go/kill decisions.
- » Risk/reward optimization and timing decisions
- » When it is questions about money there you face conflicts too. Proper conflict management is needed.
- » Investments (costs) throughout the life cycle of product not only during development phase

REVENUE / PROFIT ALLOCATION

Who owns the design?

Product consisting of
products (processes,
plants)

Transfer prices between legal units?
Taxation and local legislation?
Customs?

Who owns technologies
used in products?

Product consisting of
products

What is fair?
How to balance multi dimensional optimization, multiple KPI?
Incentives?

Manufacturing

Sales organization

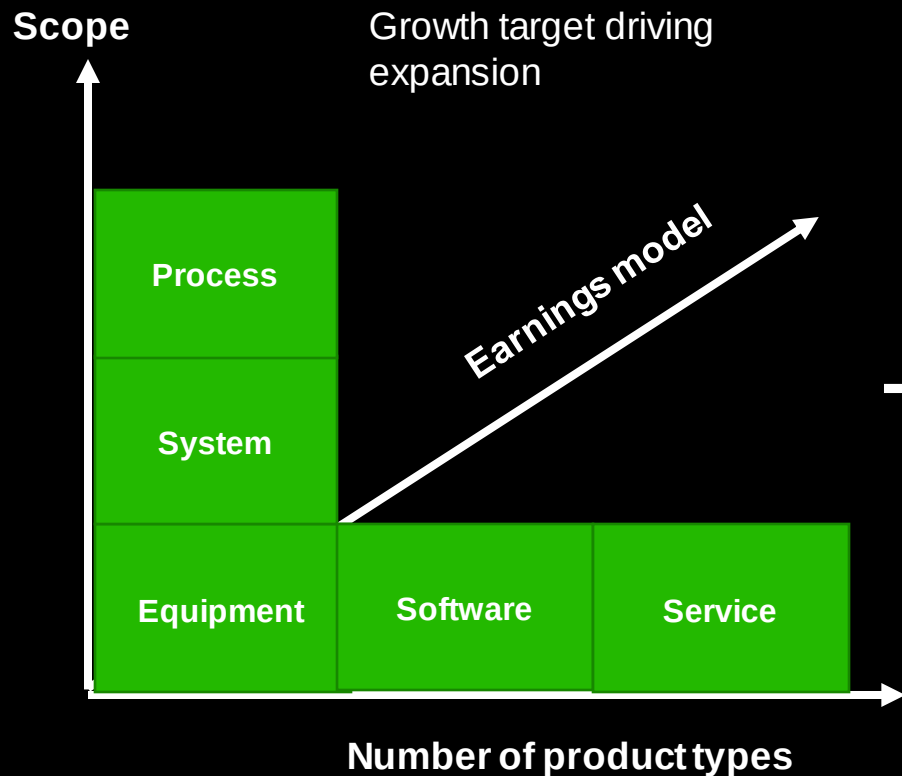
Distribution to
channel

Warranty costs

Life cycle revenue

» Multiple questions and sometime very tricky to answer

INCREASING COMPLEXITY

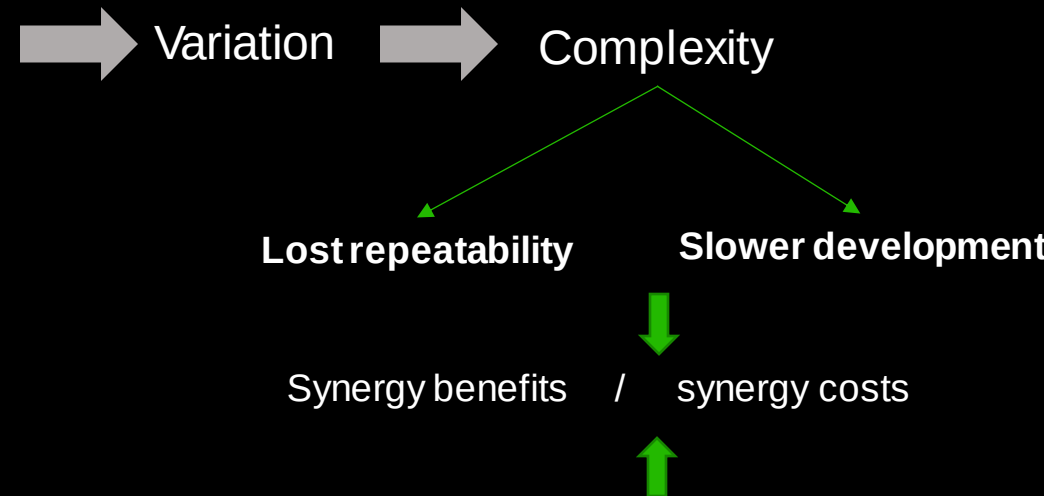


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New

- Customer types
- Distribution channel
- Supply models
- Etc.

Not actively retiring products



Consider: over 50% companies competitive edge is cost efficiency

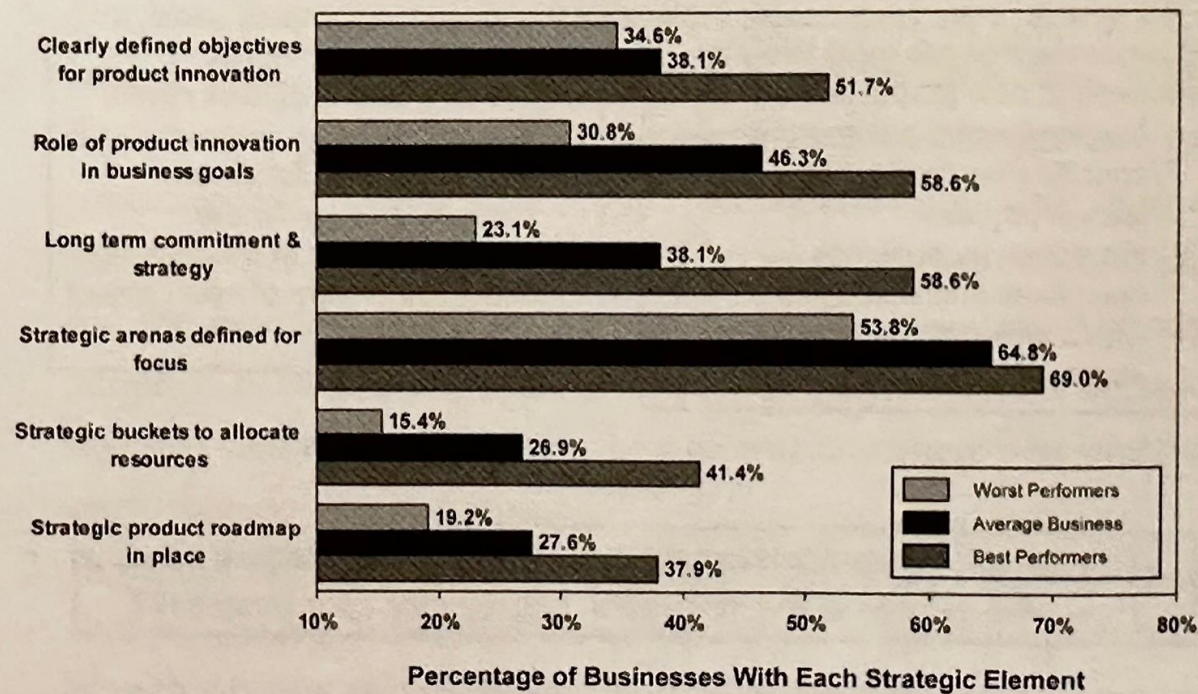
» Complexity is the silent killer



THE REALITY

Examples

THE REALITY



Typically, systematic approach is missing → Potential risks in product development

CASE WIND TURBINE INDUSTRY

- Amazing similarity among industry key players
- What went wrong?

There's no publicly available industrywide data on turbine failures, making it tough to paint a complete picture of changes in their performance over time. But Vestas and GE have said the shares of their machines in the field that are unable to produce power are elevated, even if it's still a small proportion of their installed fleets. Siemens Energy revised its earnings outlook for 2023 downward this month, citing higher-than-expected costs caused by flaws in Siemens Gamesa's installed turbines.



CASE WIND TURBINE INDUSTRY

Vestas, GE and Siemens Gamesa have confirmed in statements to Bloomberg and in recent calls with analysts that the push to rapidly develop more powerful turbines has led to challenges. The companies say they are focusing on improving manufacturing operations and have acknowledged that it's time to tap the brakes on the introduction of designs. "Rapid innovation strains manufacturing and the broader supply chain," GE CEO Larry Culp said on an earnings call in October. "It takes time to stabilize production and quality on these new products."

These days manufacturers are focused on producing machines with better reliability, at scale. In an interview this month, Henrik Andersen, the Vestas CEO, said that turbines are big enough for now, and that increasing production will be the key challenge of the next decade. Siemens Gamesa's CEO Jochen Eickholt has told investors the company is working to increase standardization among its products, to prune a portfolio that had become too broad. GE's Culp said in the October earnings call that his company is likewise shifting to "workhorse products, so we and our suppliers can implement more repeatable manufacturing processes."

Multiple
reasons for
failure

ANALYSIS

Prod. portfolio
management

Platform
management

Product
development

Supply
chain and
logistics

Production
& operations

Rapidly develop.....led to problem

Too broad portfolio

Standardization among products

Shifting to workhorse products

Repeatability in manufacturing

Better reliability, at scale
Increasing production

- **Product Portfolio management?**
- **Platform management?**
- **Horizontal alignment?**

CASE TOYOTA VS. HYUNDAI



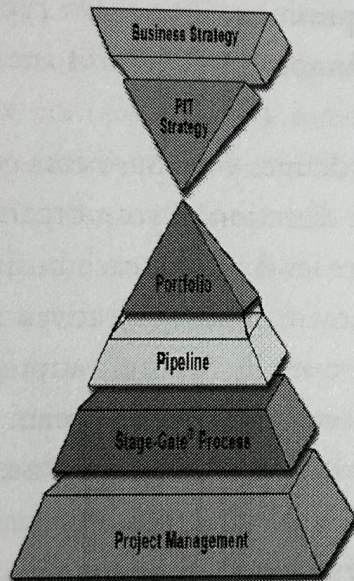
Impact of Product / technology strategy ?

"The mission of the new team, led by new President Sato, is to fully remodel Toyota as a mobility company," Toyoda stressed.

New technologies?
New business models?
A long transformation

MULTIPLE GOVERNANCE MODELS

The Innovation Pyramid



Governance

- Board of Director, CEO and Senior Leadership Team
- CEO and Senior or Corporate Leadership Team
- Subset of the Senior Leadership Team – (i.e. CTO, VPs of Marketing, Sales, Business Development, Heads of Business Units)
- Senior Gatekeepers and other key members of the Business Units or Divisions Leadership Team
- Gatekeepers - includes both Senior and Junior Gatekeepers
- Project Leader and/or Project Management Office (PMO)

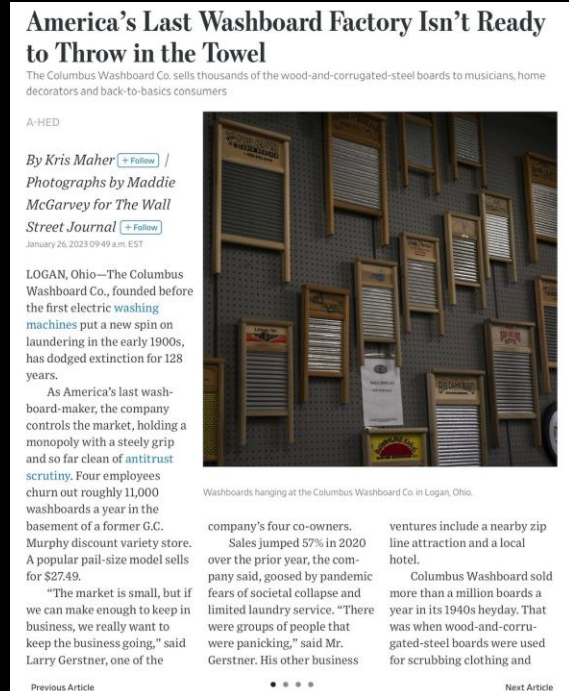
EXHIBIT 8.2: Each aspect of the Innovation Pyramid has a governance requirement.

”Einari Vidgrén kutsui puheilleen tammikuussa 2009. Mukana oli juuri uudessa tehtävässään aloittanut tuotekehitysjohtaja tohtori **Juha Inberg**. Einari sanoi: Suunnitelkaa uusi kone. Teillä on aikaa pääsiäiseen asti. Silloin pitää ensimmäisen puun kaatua. Käsky oli selvä, palaveri kesti noin kolme minuuttia.”

Uusi kone suunniteltiin ja prototyyppi valmistettiin. Yhtiö loi nykyiset kahdeksanpyöräiset lippulaivamallit ja loi myös pohjan vuonna 2013 lanseeratulle Scorpion-harvesterille. Investoinnit tuotekehitykseen ja tuoteinnovaatioihin ovat kantaneet Ponssea eteenpäin.

There's probably in excess of a dozen start-ups effectively in Tesla... Designing and making our own power electronics... manufacturing our own motors, chargers [and] the Supercharger network is a start-up. The thing, I think that people just don't really understand about Tesla is that it's a whole chain of start-ups

LIFE CYCLE CASES



GEOPOLITICS

- » Even perhaps the world most important company can't avoid geopolitics
- » Forced to maintain previous generations for China

Semiconductors

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Netherlands and Japan join US in restricting chip exports to China

Deal marks significant step in Washington's efforts to limit rival's efforts to develop its semiconductor industry



The US has been negotiating with the Netherlands and Japan for two years but faced resistance because they were worried about the effect on their chipmaking tool companies © via Reuters

By **Demetri Sevastopulo** in Washington and **Sam Fleming** in Brussels 8 HOURS AGO

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SIMPLE? IN PRINCIPLE YES, BUT MILLION WAYS TO FAIL

Assumptions

- Market: Assumptions about market and revenues were optimistic/pessimistic
- Assumption about customer needs where wrong
- Competition surprises
- Assumptions about costs were optimistic
- IP protection surprises
- Distortion Geopolitics
- Too low revenue estimates
- Changing society and values
- Rapid regulation changes

Execution

- Quality problems (under, over)
 - Design
 - Manufacturing/production
- Cost over-runs
- Over engineering
- Delays
- Subcontractor fails to meet commitments
- Matching with capabilities
- Small dirty detail

Timing

- Market timing and entry
- Development is slower
- Frequency of portfolio changes



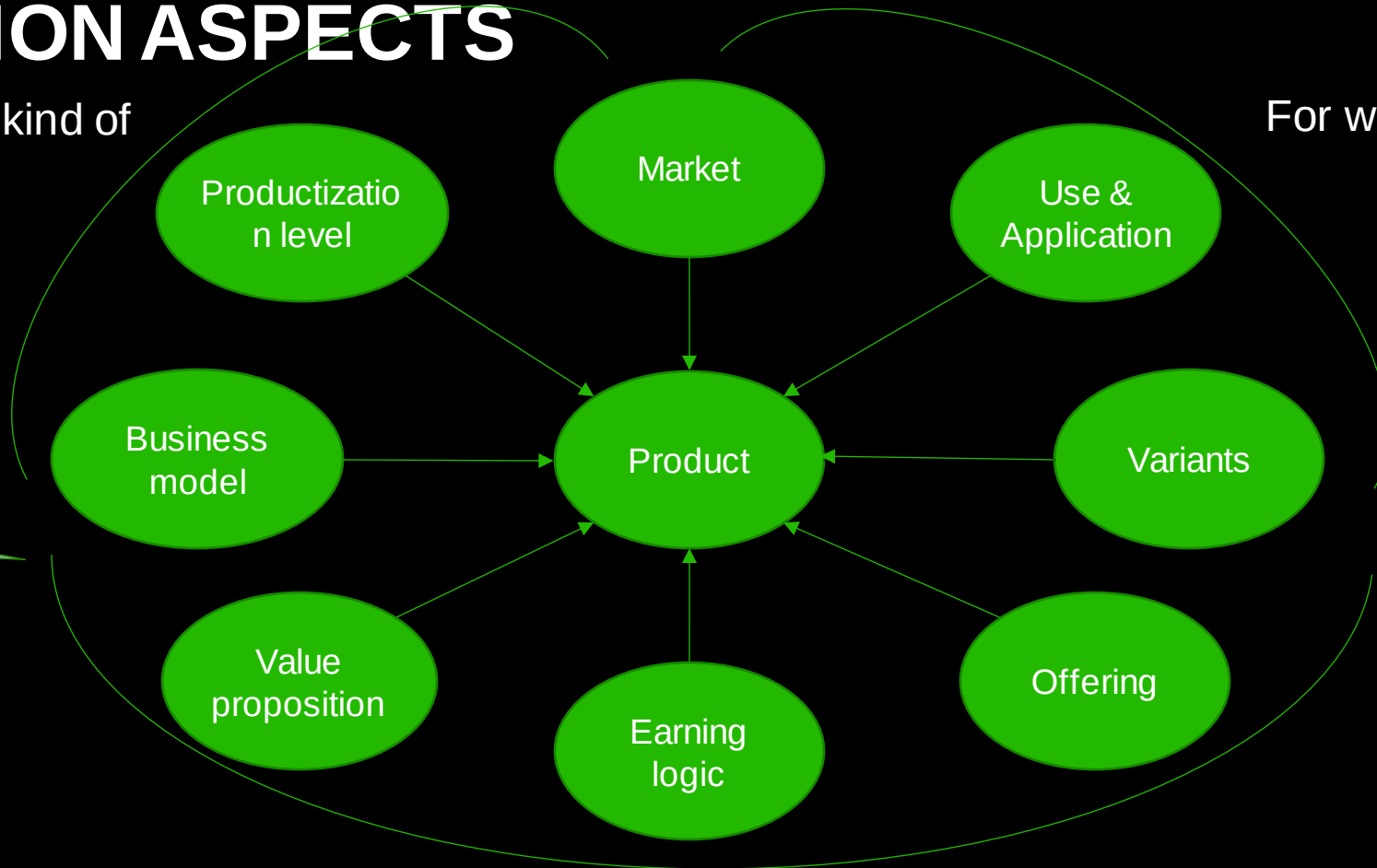
II WHAT IS OUR PRODUCT

What kind of product are we developing?

COMMON ASPECTS

What kind of

For what purpose?



Simplified

How to monetize?



PRODUCT VS. BUSINESS MODEL

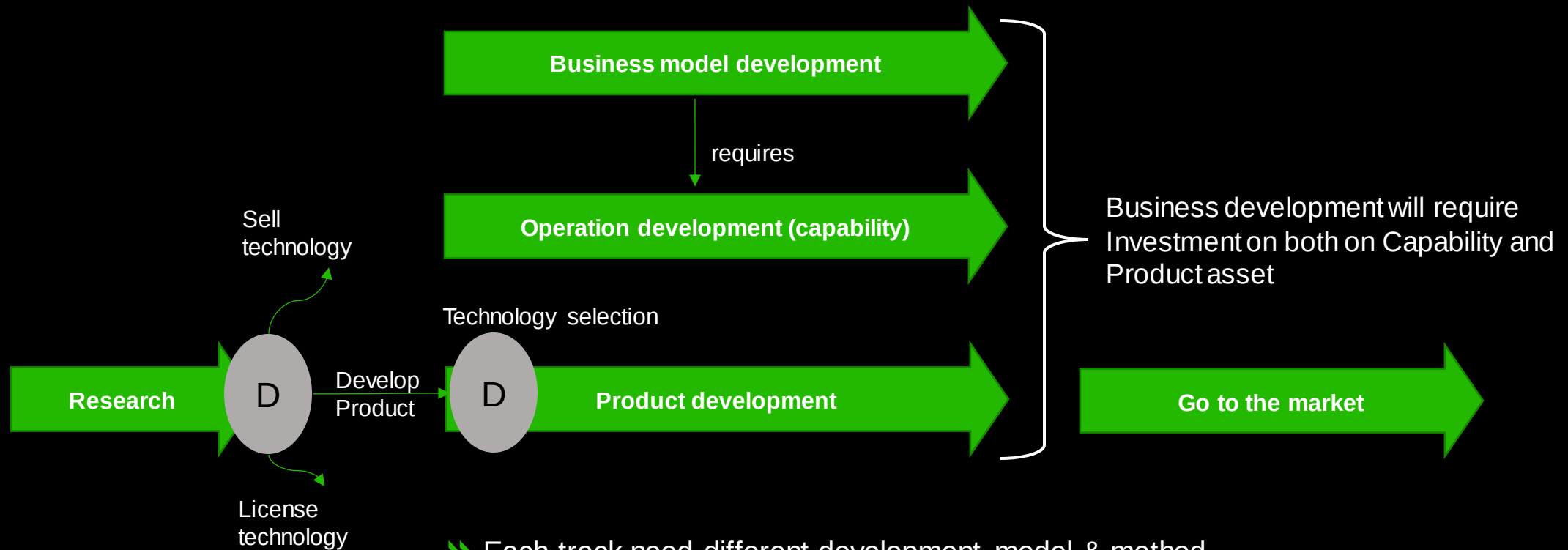
Are you developing Product or Business model or both?



BUSINESS MODEL

- » Business model is representation of the strategic choices that characterize a business venture.
- » These choices are made either intentionally or by default, so the contribution is to make them explicit.
- » Remember, at least 27 different versions of business models

SCOPING: DEVELOPING OF WHAT?



- » Each track need different development model & method
- » Need to be managed correspondently

PRODUCT VS. BUSINESS MODEL DEVELOPMENT: EXAMPLES

Business model (strategic by nature), change will require new capability

Earning model

Asset sale -> Subscription
Subscription -> Usage based
Asset sale->
Lending/Renting/ Leasing

Pricing model

Fixed pricing
• Segment
• Volume dependent
Dynamic pricing

Product

Productization level
Configuration type
New engineering discipline
Product consisting of prod.

Customers

B2B -> B2C
Customers' customers
SME -> Large Corporation

Channel

Customer service -> Online channel

Competing with price?

Case VR?

Business model change

Case new wellbeing services county

Product level development

Offering

Naming
Pricing structure
Price levels

Design Engineering

Architecture redesign
Modularization
Standardization

Manufacturing engineering

Manufacturing
Process change (BoP)
Make of buy

Delivery

Transportation packaging

» Typical changes, borders are not always that clear

EXAMPLE: MOBILE PHONE AS A SERVICE



» Governance: who is responsible of decisions?



SUMMARY

- »» You had to choose how to achieve targeted outcome
 - »» Business model development -> capability development
 - »» Product development
 - »» Technology development
- »» Mixing different development domains in the same process is challenging (Technology, Product, Business model)
- »» Mixing creates difficulties in cost and profitability calculation, too.



PRODUCTIZATION LEVEL

What is the level of invest needed to be competitive?

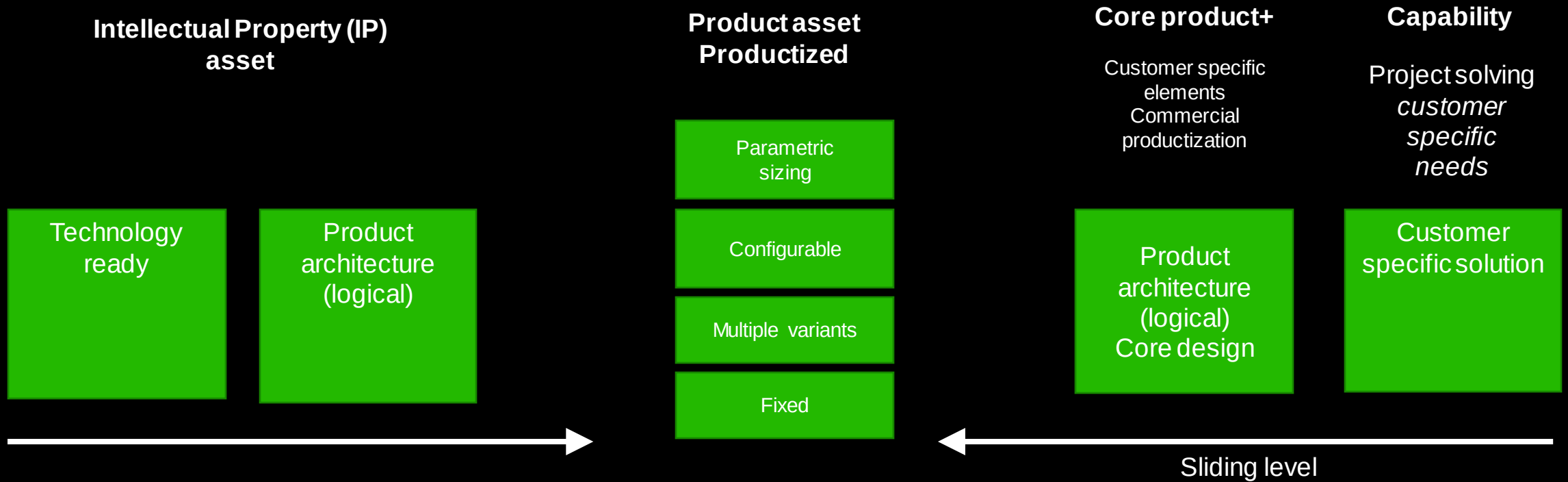


PRODUCTIZATION LEVEL : AN EXAMPLE



MARKET SEGMENT SELECTION

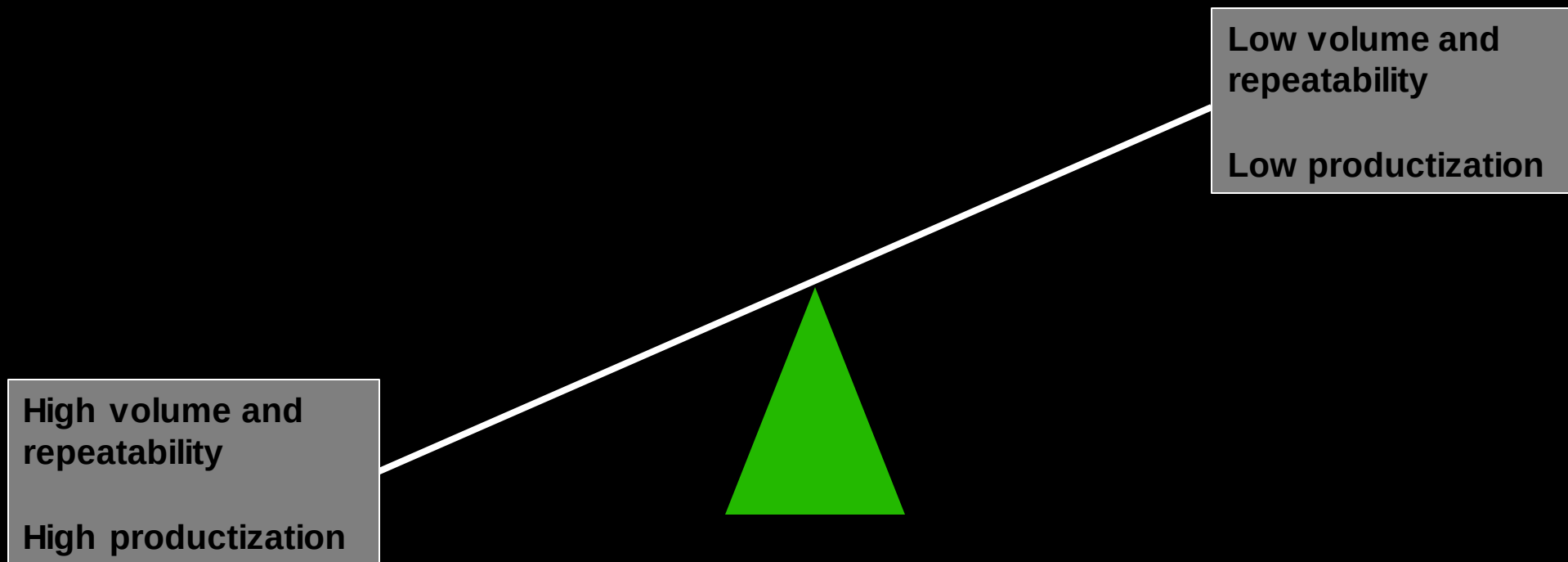
The tough selection



- » Repeatability
- » Impacts heavily on the required investment amount



PRODUCTIZATION DECISION





ANY
QUESTIONS
?

